CALIBER 2026 | BUSINESS PROPOSAL | PRELIMINARY STAGE

Asset Knowledge Platform

An AI-enabled Manufacturing Knowledge Hub that unifies plant knowledge in one governed source, answers engineering questions with verifiable evidence, and turns maintenance history into operational decisions.

Chosen use case: Case 1, Manufacturing Knowledge Hub

98	1,109	20	128/128
documents unified	searchable passages	computed AI tools	evaluation questions passed

Team: [Team name]

University: [University] **Supervisor:** [Supervisor name]



Working prototype built on the case baseline data, with a 3-minute demonstration video

Problem Analysis: Fragmented and Unverified Knowledge Is the Systemic Cause of Improper Execution Risk

A petrochemical plant depends on correct procedures, current drawings and past lessons; today these cannot be found, trusted or connected quickly

Evidence from the baseline data

98
source documents in 8 formats, held in separate repositories

48
conflicts and gaps between drawings, datasheets and interlock logic

211
work orders in 1.52 years with no link to procedures or drawings

31
breakdowns, causing 434 h of downtime

IDR 538 M
maintenance cost, of which 82% corrective

21%
of faults detected before they became corrective work

Problem statement

How can the plant connect scattered knowledge into one trusted, traceable source that engineers can use to execute, troubleshoot and improve operations without improper execution?

Sub-problems

- **Knowledge is scattered and unstructured**
- **Answers are slow to find and cannot be verified**
- **History is not used to guide action**

Root causes

- SOPs, datasheets, P&IDs, EDMS, AIMS and logs sit in separate systems
- No common asset key links a drawing, a procedure and a work order
- Revision and approval status are not visible at the point of use
- Conflicting documents go unnoticed (48 found)
- Tacit expertise is held by a few senior engineers
- Work orders, RCA and lessons are not matched to new faults
- Hazard, permit and readiness checks are separate from the asset context

Engines

- **TAGLINK** — Industrial Data Ops
- **VERA** — Verified answers
- **FORESIGHT SAFEWORK** — Decision support

Stakeholders affected

Operations	step and set point mid-shift
Maintenance	past failures, right part
Reliability	repeat failures, root cause
Process eng.	limits, design basis
HSE	hazards and permits per asset
Management	one consistent answer

Illustrative conflict: valve FV-2302 is labelled both "trips OPEN" and "trips CLOSED"; one sheet is titled KA-4501 while its body states KC-4501.

Source: CALIBER 2026 case book (Case 1); baseline data analysed in the prototype (8 equipment sets, LLDPE unit, June 2024 to December 2025)

Solution: Asset Knowledge Platform Connects Plant Knowledge, Verifies Every Answer and Puts Lessons to Work

Three layers answer the three key questions of the case (Q1, Q2, Q3); four engines run on one tag-keyed knowledge core

Main problem

48 conflicts across 98 plant documents

No single source that engineers can trust, so they search, ask colleagues or guess.

Objective

1. **Connect:** scattered knowledge in one governed, tag-keyed base
2. **Verify:** every answer with source, revision, status and confidence
3. **Act:** on failure memory, risk and hazards in daily work

Implications

Engineers	▼ Search time	▼ Wrong steps
Plant	▲ Reliability	▼ Late start-up

CONNECT Industrial Data Ops — Q1

TAGLINK engine

1. AI-ASSISTED INGESTION

SOP · Datasheet · P&ID · Interlock C&E · Maint. log · Incident / RCA · OPL

Algorithm: schema-constrained LLM extraction; each value must quote its source or it is dropped; secrets masked

2. TAG-KEYED DATA MODEL

GA-1201A → SOP rev 3 · 6 work orders · P&ID · OPL lesson

Algorithm: 1,109 passages, typed facts and relations joined on the equipment tag; chunk hash and canonical status

3. QUALITY GATES AND APPROVAL

Draft ▶ In review ▶ Approved

Algorithm: rule-based conflict detection (48 found); status trust weights from 1.00 (approved) to 0.35 (superseded)

ASK Verified answers — Q2

VERA engine

1. INTENT ROUTING

"What must be healthy before pump GA-1201A can start?"

Algorithm: intent classifier sends the question to one of 20 computed tools or to evidence retrieval

2. HYBRID RETRIEVAL

Keyword · Meaning · Trust, scope

Algorithm: k1 1.4, b 0.75; tag, status, equipment and role weighting; embedding fusion when connected

3. VERIFIED ANSWER

4 permissives: suction valve open, seal flush dP above 1.5 bar, min-flow valve open, no active trip [1]

SEQ-1201 Rev 3 · Issued for Operation · Confidence 85%

Algorithm: weighted evidence score; numbers and quotes checked against sources; bypass requests refused

ACT Decision support — Q3

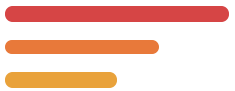
FORESIGHT and SAFEWORK engines

1. FAILURE MEMORY

WO-240088	Crosshead vibration, KC-4501
Cause	Pin clearance excessive
Action	Renewed pin and bush, set clearance

Algorithm: similar failures retrieved by symptom and cause overlap; recurrence within 90 days counted

2. RELIABILITY AND READINESS



Algorithm: failure-mode RPN; gamma-Poisson 90-day forecast with ranges; backtested score; GO / NO-GO

3. PLANNING AND INTEGRATION

EDMS · AIMS · Digital twin · Historian

Algorithm: permit-aware four-week plan with clash rules; hazard screening models; read-only system links

CORE TECHNOLOGIES

- BM25F + embeddings
- LLM with tool calling
- Typed fact graph
- Bayesian forecasting
- Hazard screening models
- Serverless, Postgres next

Source: Prototype built on the case baseline data; counts and parameters are taken from the running application

Business Impact: IDR 1.5 B Annual Value in the Base Case, Payback in About 21 Months, With the Downside Disclosed

Financial, operational and people benefits with measurable KPIs; hours and rates are assumptions to be replaced by Company figures during the pilot

Financial assessment

Parameter	Baseline	With platform
Time to a verified answer	about 15 min*	3 min
Mean time to repair (MTTR)	14.0 h	12.6 h
Repeat failures within 120 days	8 of 55	5.6 of 55
Annual benefit (IDR M)	0	1,521
One-off investment (IDR M)	0	1,400
Annual running cost (IDR M)	0	370

IDR 1.44 B
net benefit over 3 years

40%
internal rate of return

IDR 0.91 B
net present value at 10%

21 months
payback period

Low case
IDR 0.8 B a year
no payback in 3 years

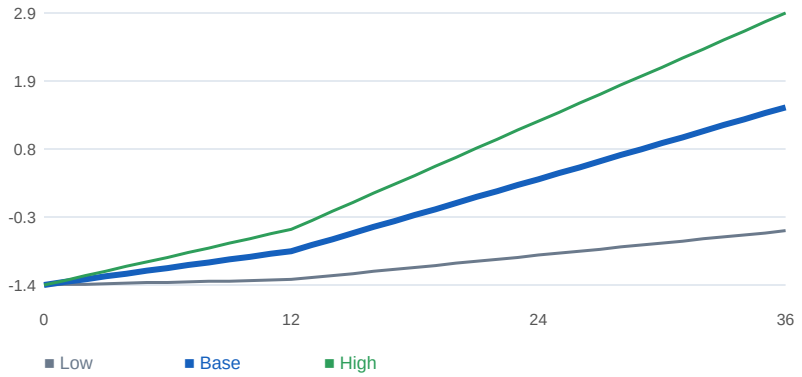
Base case
IDR 1.5 B a year
payback 21 months

High case
IDR 2.1 B a year
payback 16 months

Annual value by lever, base case (IDR M)



Cumulative net benefit (IDR B)



Operational, safety and people impact

Execution quality
100% of answers carry source, revision, approval status and confidence

Reliability
MTTR down 10%, repeat failures down 30%, early detection from 21% to 35%

Process safety
Interlock bypass requests refused; hazard, permit and readiness checks at the point of work

Knowledge retention
56 approved lessons today, plus 12 each quarter, searchable by asset

Data integrity
48 document conflicts owned; 90% closed within 6 months

Onboarding
New engineers productive one month sooner (4 hires a year assumed)

* assumed; all other baseline values come from the 211 work orders

Source: Case baseline data (work orders, MTTR, breakdowns); assumptions in appendix A16. IDR = Indonesian rupiah; IRR and NPV over 3 years, 10% discount rate

Feasibility and Roadmap: Read-Only, Advisory Deployment in Four Phases With Explicit Decision Gates

Readiness across technical, operational, legal, cybersecurity, data governance and organisational dimensions, and a phased plan from the November 2026 finals

Technical
Ready
- Prototype runs end to end on the case data
- SQL schema and API specification written
- Model provider is replaceable by contract

Operational
Ready
- Advisory only; no write access to DCS
- Local search continues if the model is offline
- Fits existing OPL and work-order routines

Legal
Decision needed
- Enterprise AI terms, no training on Company data
- Data held in region (PDP Law)
- Drawing and vendor IP respected

Cybersecurity
Planned
- Hub kept outside the OT zone (IEC 62443)
- Roles, audit trail and secret masking
- Prompt-injection guards and rate limits

Data governance
Ready
- Owner, revision and status on every record
- 48 conflicts held in a managed register
- Approval before knowledge is trusted

Organisation
Planned
- A champion in each function
- Two-hour role-based training
- Monthly KPI review

Implementation roadmap

	Q4 26	Q1 27	Q2 27	Q3 27	Q4 27	Q1 28	Q2 28	Q3 28	Q4 28
0 Mobilise	Start	Finals 4-5 Nov 2026: data access, owners, enterprise AI terms							
1 Pilot, LLDPE unit		Pilot	90-day pilot: 45 users, case data live, success gates						
2 Connect			Integrate	EDMS and AIMS adapters, single sign-on, security review, live tag values					
3 Extend		Predictive and HSE on live data, digital-twin link, units 2 and 3			Extend				
4 Scale					Plant-wide roll-out and continuous improvement			Scale	
Decision gates		G1 Pilot gates met		G2 Security and integration sign-off		G3 Value review		G4 Scale decision	

One-off budget, IDR 1,400 M

- Integration 650
- Data 350
- Change 250
- Security 150

Run cost IDR 370 M a year: hosting, model use, monitoring and half a support role

Wrong answer: evidence-only drafting, verification, refusal rules

Poor data quality: conflict register, named owners

Low adoption: champions, tools used in daily work

Model cost or outage: replaceable model, offline fallback

Source: Prototype and architecture notes; dates and budget are proposals for discussion with the Company

Conclusion: A Trusted, Traceable Knowledge Hub Turns Scattered Plant Knowledge Into Faster, Safer and More Reliable Decisions

Summary of the proposal, how it answers the three key questions of the case, and the decision requested

CONNECT TAGLINK: one governed base

- 98 documents and 211 work orders unified on the equipment tag
- 48 conflicts detected and assigned to owners
- Ingestion, quality gates and approval for every record

98 to 1 documents into one governed base

ASK VERA: answers with evidence

- Source, revision, approval status and confidence on every answer
- Numbers and quotes verified against the cited text
- Safety bypass requests refused with the reason

128/128 evaluation questions passed

ACT FORESIGHT, SAFEWORK

- Failure memory with RCA and lessons at the point of fault
- Risk ranking, readiness, hazard screening and permit-aware planning
- Read-only links to EDMS, AIMS and the digital twin

20 tools computed from plant records

How the proposal answers the three key questions of the case

Q1	**Build a structured Industrial Data Ops foundation**	Tag-keyed data model over seven source types; AI-assisted ingestion, quality gates and approval with named owners; conflicts surfaced on day one	**Evidence:** 98 documents, 1,109 passages, 48 conflicts registered
Q2	**Help engineers find trusted answers faster and reduce improper execution**	Hybrid retrieval, evidence-weighted confidence score, verification of numbers and quotes, refusal of unsafe requests	**Evidence:** 128/128 questions passed; 3 minute target versus about 15 minutes today
Q3	**Integrate with operational systems for reliability and improvement**	Read-only EDMS, AIMS, historian and twin links; failure memory, backtested reliability analytics, readiness, hazard screening and site planning	**Evidence:** 64-forecast backtest; 20 computed tools

Decision requested: a 90-day pilot

Weeks 1 to 2
Agree data access, owners and AI terms

Weeks 3 to 6
Load the LLDPE unit and train 45 users

Weeks 7 to 12
Measure against the success gates, then decide on scale

Source: Prototype test results (appendix A17) and business case (slide 4, appendix A16)

Team Profile

Team name, university, supervisor and members (replace the bracketed fields before submission)

Team name: [Team name] **University:** [University] **Supervisor:** [Supervisor name]

No.	Name	Major	Semester	Area of expertise	Contribution to the proposal
1	[Name]	[Computer Science]	[6]	[AI and software]	[Formulated the idea; built the prototype and the AI assistant]
2	[Name]	[Chemical Engineering]	[7]	[Process and reliability]	[Problem analysis, process content and hazard models]
3	[Name]	[Industrial Engineering]	[7]	[Maintenance and business case]	[Business impact, KPIs and roadmap]

The booklet allows two to three members from the same university. Delete unused rows.

A1 Platform Feature Map: Twenty-Four Capabilities Across Four Functional Groups

Every feature runs on the same tag-keyed knowledge core and is available in the working prototype

Industrial Data Ops (Q1)	Verified answers (Q2)	Decision support (Q3)	Plan and HSE (Q3)
Ingestion wizard Nine document kinds, column mapping, AI pre-fill	**AI assistant** Streaming answers, offline fallback	**Failure memory** Similar failures, causes, actions, recurrence	**Hazard simulator** Cloud, jet, pool, blast, muster walk
Quality register 48 conflicts with owner and severity	**20 computed tools** KPI, interlock, hazard, forecast, spec	**Reliability analytics** RPN, MTBF, MTTR, forecasts, backtest	**Maintenance site planner** Four-week plan, permits, clashes
Approval workflow Draft, review, approved, superseded	**Hybrid search** BM25F with embeddings when connected	**P&ID with valve logic** Closing a valve blocks flow and trips	**Plant replay** The plant on any date of the history
Typed facts and links 447 facts, 624 relations on the tag	**Source panel** Revision, status and score per source	**Interlock and OPL simulators** Cause and effect on the machine in 3D	**3D plant and equipment** Selectable parts, live state, 8 models
Roles and audit trail Role filter, change history, masking	**Confidence and checks** Evidence score, number and quote checks	**Start-up readiness** GO, GO WITH CHECKS, NO-GO	**Report Center** Reliability, readiness, KPI exports
Knowledge Studio Capture of tacit lessons and RCA	**Feedback and evaluation** Useful, incomplete, wrong; 128 test questions	**Chain-reaction scenarios** Which units trip next, for how long	**Performance dashboard** Availability, downtime, cost trends

Source: Prototype feature inventory

A2 System Architecture: Five Layers on One Tag-Keyed Knowledge Core

Data flows from plant sources to users, with governance applied to every layer

1 Sources	2 Industrial Data Ops	3 Knowledge core	4 AI services	5 Experiences
SOPs, OPLs, tacit notes	AI document ingestion	Documents and passages	Intent router	AI assistant
Datasheets	Structure and tag mapping	Search index (BM25F, vectors)	Retrieval and ranking	Equipment 360 in 2D and 3D
P&ID, GA, plot plan	Quality gates, conflicts	Typed facts: tags, set points	20 computed tools	P&ID, interlock, OPL tools
Interlock logic (C&E)	Secret masking	Relations between records	LLM drafting, tool calls	Maintenance, failure memory
Maintenance logs, RCA	Owner, revision, status	Failure memory, lessons	Verification, confidence	Site planner and replay
Incident reports	Approval workflow	Answer, audit, feedback log	Reliability, hazard models	Hazard and readiness
Live data (DCS, historian)				

Cross-cutting governance: role-based access | approval before trust | audit trail | prompt-injection guards | retention and region | read-only toward plant systems

Prototype today: Single-file web application, Netlify functions and Blobs, free Gemini model, local fallback search

Target production: Postgres with pgvector, ingestion queue, single sign-on, EDMS and AIMS adapters, Company-approved model

Source: Asset Knowledge Platform architecture notes (docs/ARCHITECTURE.md)

A3 Industrial Data Ops: Six-Step Ingestion From Raw Document to Governed Knowledge

TAGLINK engine: every record is parsed, structured on the equipment tag, checked and approved before it is trusted

1 Parse

CSV delimiter detection (comma, semicolon, tab, pipe) with quoted fields; text extraction from documents

2 Classify

Nine kinds: SOP, datasheet, P&ID or drawing, interlock C&E, maintenance log, instrument list, OPL, incident report, notes

3 Extract

Server masks secrets and personal numbers, requests JSON from the model, then drops every step, value or fact that does not quote the source text

4 Normalise

Equipment tag pattern, units and operators (<, >, set point) parsed into typed facts; statuses mapped to one vocabulary

5 Check

Rule-based conflict and gap checks; a person reviews the pre-filled result

6 Approve and index

Draft, In review, Approved; item, facts and source copy are chunked, hashed and added to the index

Status trust weights used in ranking



Conflict and gap rules

- Title tag differs from the tag in the sheet body
- Duplicate or reused tag for different valves
- Valve fail position stated both open and closed
- Drawing reference missing or a placeholder
- Revision shown without a date
- Dimensions disagree between datasheet, GA and plot plan
- Trip description inconsistent with the instrument type
- Result: 48 findings, each with severity, source and owner

Resulting knowledge base

1,109
searchable passages

447
typed facts

624
relations on the tag

98
documents, 84 instruments

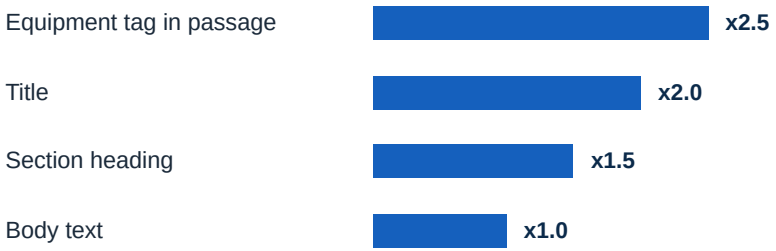
Source: Prototype ingestion wizard, extraction service and data layer (dl.js, ingest.js, wiz.js, facts.js)

A4 Retrieval: BM25F Ranking With Tag, Status, Equipment and Role Weighting

VERA engine, step 1: the best evidence is selected before any language model is used

1 Score each passage

BM25 with field weights: a term counts more when it appears in a heading or in an equipment tag.



k1 = 1.4, b = 0.75

Query tags weighted 2.2; synonyms and stemming applied; broad words down-weighted

Why BM25F: plant text is short, tag-heavy and full of identifiers that exact matching handles well. Embeddings add recall for questions that share no words with the passage.

2 Re-weight by trust and context

Status trust	0.35 to 1.00
Source kind prior	0.55 to 1.05
Equipment in scope	x 1.45
Other equipment	x 0.70
Role fit (operator, HSE, reliability, manager)	x 1.10 to 1.15

Source-kind priors: interlock 1.05, OPL 1.05, document 1.00, work order 1.00, plot plan 0.95, instrument 0.85, data-quality note 0.55

Access control

Passages carry an access list by role. Draft and superseded items are visible to staff only; rejected items are never returned.

3 Add meaning and diversify

Embedding fusion (when connected)

- score = (1 - 0.35) x BM25F + 0.35 x cosine x 40
- Semantic recall: a passage with no shared words qualifies when cosine is at least 0.6

Diversity rule

- At most two passages per source document, so one long file cannot fill the context

Output to the answer stage

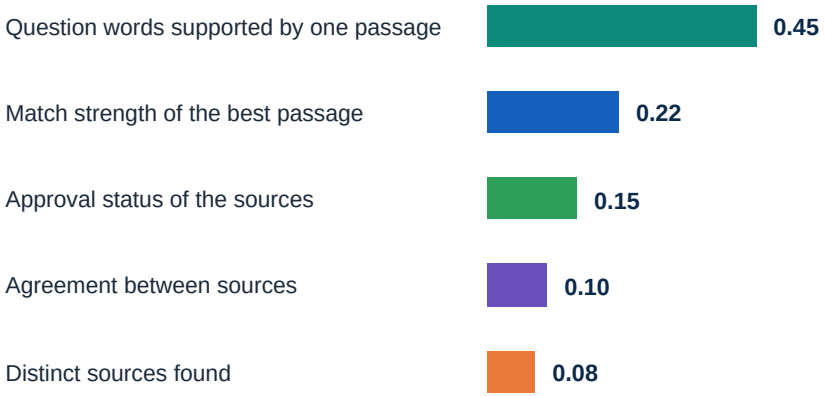
- Ranked passages with chunk id, revision, status and matched terms
- Relative score for each passage
- Question words covered, used by the confidence score

Source: Prototype retrieval index (dl.js); parameters are the values used in the running application

A5 Answer Generation: Evidence-Weighted Confidence, Verification and Safety Refusals

VERA engine, step 2: the model words the evidence; the platform measures how well the evidence supports the answer

Confidence score (0 to 100)



Caps applied after weighting

- Broad question without a specific term: maximum 60
- Sources are for different equipment: maximum 40
- Fewer than half of the question words in the sources: maximum 45
- No single passage covers the question: maximum 50

High 70 or more | Medium 45 to 69 | Low below 45

Calculated tools report their own confidence, because their figures come directly from records.

Pipeline and checks

1. **Intent routing**
   Classifier selects a computed tool or retrieval
2. **Tool or evidence**
   Tools return figures; retrieval returns passages
3. **Drafting**
   Model writes only from numbered passages
4. **Number check**
   Every figure with a unit must appear in a source
5. **Conflict and status notes**
   Open conflicts and draft sources are flagged on the answer
6. **Logging**
   Question, sources, score and latency stored

Twenty computed tools

Reliability
maintenance_stats, kpi_summary, failure_modes, predictive_maintenance, failure_memory, history_lookup, spares_outlook

Assets and limits
equipment_info, spec_lookup, tag_facts, plot_plan_query, compare_equipment, rank_equipment

Operations and safety
interlock_info, procedure_help, consequence_analysis, cascade_analysis, hazard_effects, safety_guard

Data governance
data_issues

Safety refusal: a request to bypass an interlock is declined with the reason and the approved route.

Source: Prototype assistant (ai.js, intent.js, ai2.js); weights and caps are the values used in the running application

A6 Failure Memory: Similar Failures, Causes and Actions Retrieved at the Moment of Need

FORESIGHT engine: every new symptom is matched against work orders, one point lessons and approved user lessons

1 Describe	2 Retrieve	3 Parse	4 Group	5 Check recurrence	6 Recommend
Symptom text and optional equipment or component	Retrieval over work orders, OPL troubleshooting cases and approved lessons	Symptom, cause and action fields; component taken from the part list	Causes and actions merged by a key of the first four meaningful word stems	Same component failing again within 90 days	Ranked actions with case counts, recurrence rate and downtime range

Worked example from the case data

Fault	Crosshead vibration approaching trip on the recycle gas compressor KC-4501
Matched	Work order WO-240088, plus OPL cases on KC-4501 trips
Probable cause	Crosshead pin clearance excessive
Recommended action	Renew crosshead pin and bush, set the clearance, confirm vibration returns to normal
Recurrence	Counted when the same component fails again within 90 days
Output	Case links open the full work order, drawing or lesson

Closing the loop with Knowledge Studio

Draft lesson generation

- When a failure mode has no lesson, a draft lesson is built from its work orders
- An engineer reviews and approves it

Lessons feed retrieval

- Approved lessons are indexed immediately
- They appear beside the next matching fault

Lesson coverage

- Each failure mode shows whether an OPL covers it
- Uncovered modes are flagged as gaps

Limits

- Matching is lexical with stemming, not semantic, unless embeddings are connected
- Quality depends on how well work orders are written

Source: Prototype failure memory (fm.js, pdm2.js)

A7 Reliability Analytics: Failure-Mode Risk, Bayesian Forecasts and a Backtested Score

FORESIGHT engine: quantitative models built on 211 work orders, validated by rolling-origin backtesting, with limits stated

1 Failure-mode risk priority

- Mode taxonomy read from problem, cause and action text (10 modes)
- RPN = Severity x Occurrence x Detection
- Severity: percentile of downtime (55%) and cost (45%), scaled 1 to 10
- Occurrence: 1 + 2.4 log2(1 + 3 x rate per year), maximum 10
- Detection: 1 + 9 x (1 - share found early), minus 1.2 if an OPL covers the mode

2 Gamma-Poisson mode forecast

- Prior from the fleet rate: shape 1.5, rate 1.5 / fleet rate
- Posterior after n own events in T days: shape 1.5 + n, rate prior + T
- P(failure in h days) = 1 - exp(-mean rate x h)
- 80% credible range from the gamma quantile (Wilson-Hilferty)
- Small samples shrink toward the fleet instead of over-reacting

3 Measures and trend

- MTBF = (operating hours - downtime) / breakdowns; MTTR = downtime / breakdowns
- Laplace trend test: u above 1.64 means failures are accelerating, below -1.64 slowing
- Share of corrective jobs that repeat the previous mode
- Health index from recent breakdowns, preventive-maintenance gap and trend

4 Combined unit score and its validation

- Ridge-regularised logistic regression on eight standardised features: recent breakdowns, time since last breakdown against the usual gap, trend, repeat failures, overdue preventive work, lesson warning signs, recent corrective jobs, low health
- 300 gradient-descent iterations with an L2 penalty
- Rolling-origin backtest: train on the past, forecast the next 90 days, move forward every 20 days
- Calibration slope shrinks the output toward the base rate

64
forecasts tested

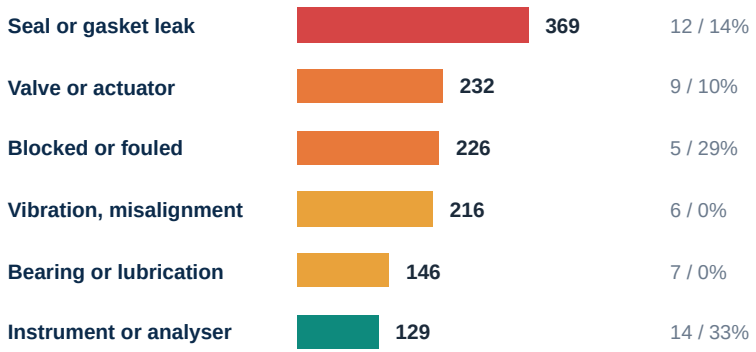
0.26
Brier score, plant-wide rate

0.47
ranking skill (AUC); 0.50 is chance

A unit's own history does not predict which unit fails next on this data. The page therefore ranks units by consequence and reports the combined score as advisory.

Failure modes by risk priority

Mode, RPN (bar), jobs, share found early



Source: Prototype reliability analytics (pdm.js, pdm2.js); 63 breakdown and corrective jobs behind the failure-mode view

A8 HSE and Readiness: Screening Hazard Models and Permissive Logic on the True-Scale Plot Plan

SAFEWORK engine: consequence screening and start-up checks use the real equipment coordinates, interlock logic and live state

1 Release	2 Source term	3 Dispersion	4 Ignition and effects	5 People and response
Orifice flow through a 10 to 100 mm hole at the operating pressure and temperature; isolation by automatic, manual or failed shutdown	Flash fraction and pool evaporation for liquids; inventory limits the release time	Gaussian plume moving with the wind, with transport delay, urban Pasquill sigmas and a dense-gas factor	Flash fire at 25, 50 and 100% of the lower flammable limit; jet and pool fire radiation; Kinney-Graham blast overpressure	Gas detection and shutdown timing; receptors in each zone; walking time to each muster point

Fluids and limits

- n-hexane, ethylene and hydrogen with their flammability limits and heats of combustion
- Heat flux zones: 37.5 kW/m2 equipment damage, 12.5 kW/m2 fatal in about a minute, 4.7 kW/m2 pain in 15 seconds
- Blast zones: 35, 21 and 7 kPa
- Pasquill stability classes A to F selectable with wind speed and direction
- Output includes release rate, cloud reach and time to the control room

Start-up readiness logic

- Each permissive from the interlock document is read against the live model
- A latched trip or a closed valve is a blocker
- Open lessons and earlier start-up problems are listed for the unit
- Verdict: GO, GO WITH CHECKS or NO-GO; sign-off recorded with a handover note
- Draft lessons are generated for modes without coverage

Intended use and limits

- Ranks and compares scenarios, trains operators and supports siting and permit decisions
- Point-source flame model and screening plume; no obstacles or terrain
- Does not replace a quantitative risk or consequence study
- The assistant calls the same model through the hazard_effects tool and states its limits

Source: Prototype hazard simulator and readiness module (hse.js, hse3d.js, ready.js); screening estimates, not a consequence study

A9 Maintenance Site Planner and Replay: Permit-Aware Scheduling With Hazard-Based Clash Rules

SAFEWORK engine: routine and forecast jobs are placed on crew days, checked against neighbouring work and shown on the plot plan

1 Generate jobs	2 Infer permits	3 Schedule	4 Check clashes	5 Show the day
Routine: median interval of past jobs per unit, type and discipline. Condition checks: top failure modes from the gamma-Poisson forecast	Hot work, confined space, lift, isolation and line break read from the wording of similar work orders	Greedy placement by due day and priority, levelled within a three-day window, within crew capacity per discipline	Pairwise rules on overlapping jobs using plot plan distances	Crane pads, laydown areas, separation rings and permits on the plot plan and in 3D

Clash rules

Hot work near an open line	separation = larger of the user setting (25 to 80 m) and twice the gas hazard radius of the unit being opened
Confined space entry	no hot work or line break in the same unit at the same time
Heavy lift	one crane: two lifts may not overlap unless a second crane is available
Crew capacity	load per discipline per day must not exceed crews x 8 hours
Lateness	a job placed more than five days after its due day is flagged

Plant replay: never miss a breakdown

- Unit state at date t: Breakdown, Maintenance, Notified or Running, from the work order start and finish times
- Playback steps are larger than most breakdowns, so the clock lands inside each breakdown before it continues
- Pause on breakdown opens the failure view for that unit; overlapping breakdowns are linked

31 / 31
breakdowns shown on the correct unit, plot plan and 3D

5
permit types checked

28 days
planning horizon with crew levelling

Source: Prototype maintenance site planner and plant replay (plan.js, rp.js)

A10 Simulation and Visual Engine: Interactive 3D and Valve-Aware P&ID Logic Without Plug-Ins

A purpose-built renderer keeps every model selectable, consistent with the drawings and responsive on ordinary laptops

3D renderer

- Software depth-buffered rasteriser on a canvas; smooth normals, outlines, transparent clouds
- Eight equipment models built from the datasheet bill of material; each part is selectable by picking
- Adaptive resolution targets a 15 ms frame; high-resolution redraw when idle

P&ID and interlock logic

- Valves open and close; closing a valve blocks flow downstream and trips the permissive that depends on it
- Interlock cause and effect evaluated from the instrument values
- Plant cascade: connectivity graph shows which units trip next and the estimated outage

Hazard, replay and planner views

- Gas cloud, flames and blast drawn at true scale on the site model
- Same state model drives the plot plan and the 3D plant
- Labels, cards and tickers are generated from the same records as the tables

Performance engineering

15 to 50 fps

equipment 3D after duty-cycled loops and visibility culling

15 to 51 fps

dashboard plant after scene caching

5%

main-thread load of the equipment viewer at rest

0

plug-ins or external 3D libraries

Why it matters: operators need the machine, the P&ID and the lesson on the same screen without installing software. Rendering is paused when a view is off screen or the tab is hidden, which keeps the platform usable on shared control-room computers.

Source: Prototype rendering engine (g3d.js, g3r.js, plant3d.js, pidv.js, scn.js); frame rates measured in a headless browser

A11 Data Sources: Each Source Has a Purpose, an Owner and a Business Justification

Baseline data from the case plus proposed additions; a small set of well-governed sources is preferred to many weakly connected ones

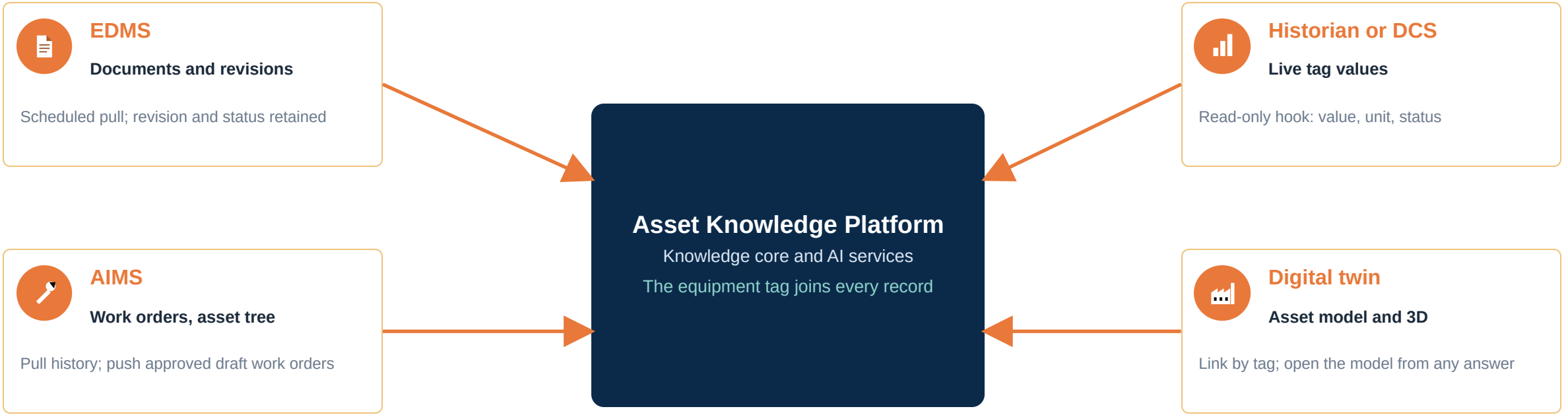
Source	Content	Business justification	Status
SOPs and OPLs (56)	Steps, safety notes, tacit lessons	Fewer wrong steps; retains SME experience	Loaded
Datasheets (8)	Design limits, materials, ratings	One place for trusted specifications	Loaded
P&IDs, GA drawings, plot plans (24)	Equipment, tags, layout, hazards	Links every tag to its drawing and location	Loaded
Interlock logic (8 sets, 84 instruments)	Causes, effects, permissives	States what trips and why, with the exact set point	Loaded
Maintenance history (211 work orders)	Failures, root causes, cost, downtime	Failure memory, KPIs, forecasting, planning	Loaded
Incident and RCA reports	Lessons learned	Prevents repeat events	Add in pilot
EDMS	Controlled document revisions	Single source of truth for revisions	Adapter specified
AIMS	Work orders and asset hierarchy	Live history; draft work orders returned for approval	Adapter specified
Historian or DCS values	Live tag values	Context for troubleshooting and prediction	Hook in place
Digital twin	Asset model and 3D	Navigation by equipment and place	Planned

Each additional source requires an owner, a refresh rule and a stated decision it improves. The case asks for a well-justified data architecture rather than more data.

Source: Case book data table; prototype counts

A12 Integration: EDMS, AIMS, Digital Twin and Plant Data Feed One Core, Read-Only First

One contract for every source: connecting a new system requires one adapter and no change to the knowledge core



Single sign-on
Roles taken from the company directory

Read first, write later
Writes are drafts approved by a person

One contract per source
A new source is one adapter

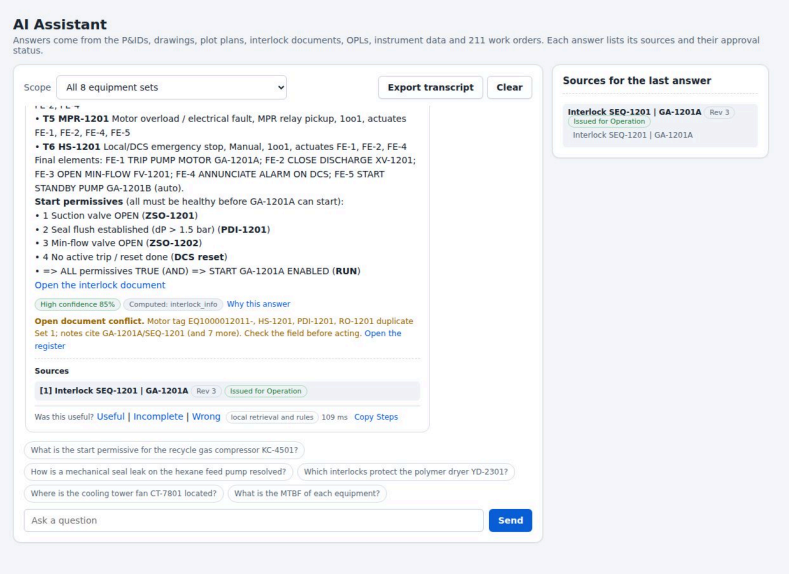
Data stays in the tenant
Region and retention set by the Company

Source: Architecture notes; adapters are specified and not yet connected to Company systems

A13 Prototype Tour 1: Ask, See the Plant, Open the Equipment

Assistant, plant overview and equipment in 3D

1 Verified answer



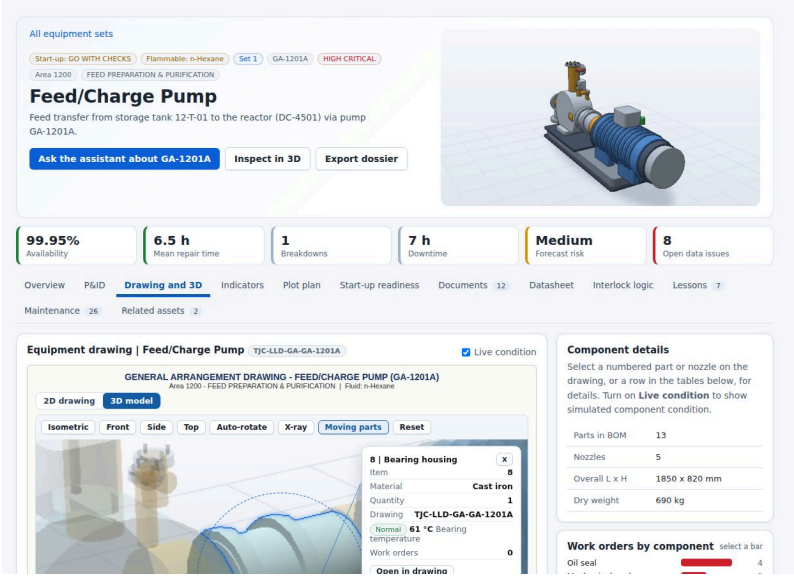
Question about start permissives. The answer lists each permissive with its tag, the source (SEQ-1201 Rev 3, Issued for Operation), a confidence level and a warning that an open document conflict touches this equipment.

2 Plant overview in 3D



Eight units on the plot plan with live state. The attention strip summarises latched trips, units not ready to start, overdue preventive work and high-severity document conflicts.

3 Equipment in 3D



The equipment drawing as a 3D model. Selecting a part shows its specification, rating and the work orders that mention it. Part names match the datasheet and the P&ID.

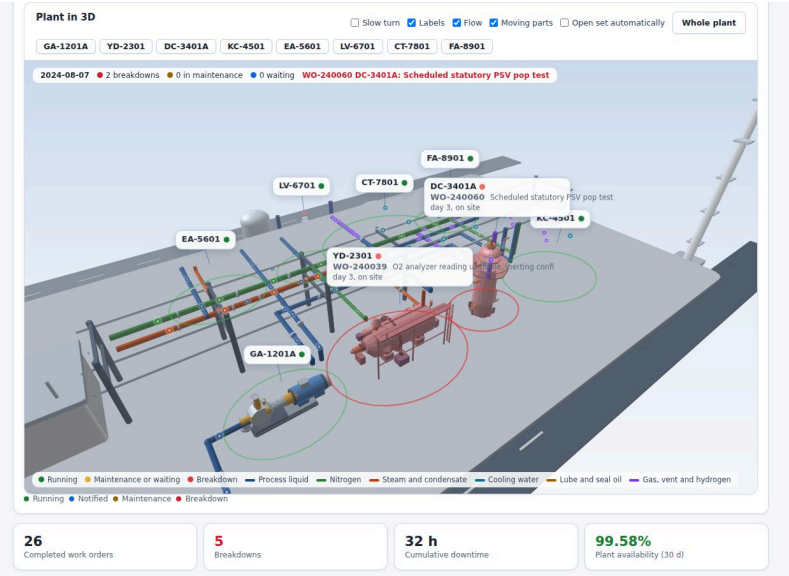
Relevance: Key questions Q2 and Q3: trusted answers found quickly, and knowledge presented in the context of the asset and its operating state.

Source: Screens from the running prototype with the case data loaded

A14 Prototype Tour 2: Maintenance Replay, Site Plan and Failure Outlook

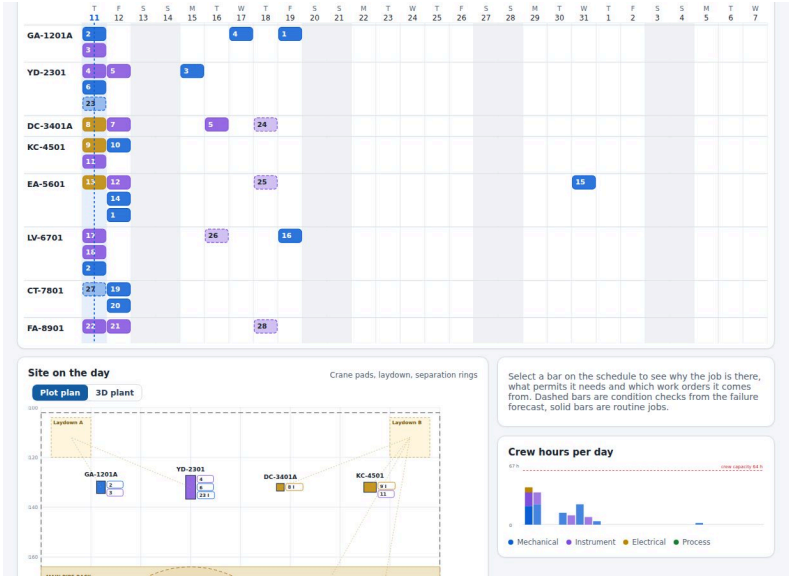
Where maintenance history becomes decisions

1 Plant replay



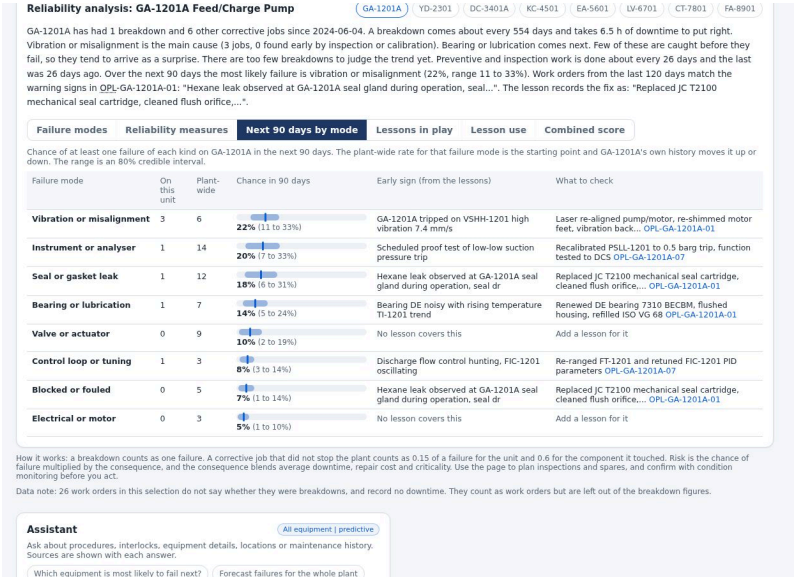
The 3D plant on any date in the history. Each unit shows its open work order and problem; a breakdown pauses the replay on the correct unit and opens its failure view.

2 Site plan



A four-week plan from routine intervals and forecast checks, with crews, permits and clashes. Crane pads, laydown areas and separation rings appear on the plot plan.

3 Failure outlook



Failure modes ranked by risk priority, the probability of each in 90 days, matching lessons and spare parts. The page states when the model cannot separate units.

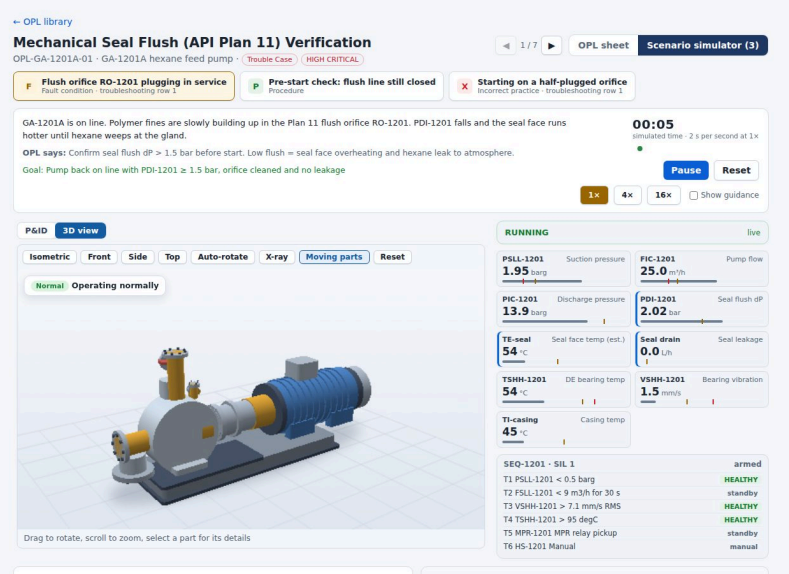
Relevance: Key question Q3: maintenance records become failure memory, a plan and a forecast that engineers can act on.

Source: Screens from the running prototype with the case data loaded

A15 Prototype Tour 3: Operator and HSE Tools on the Same Data

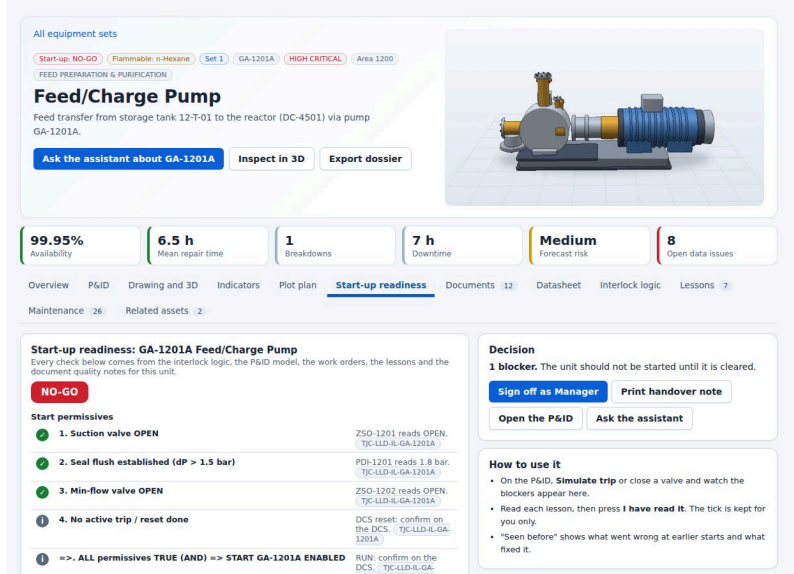
Simulators, readiness and hazard screening

1 OPL scenario simulator



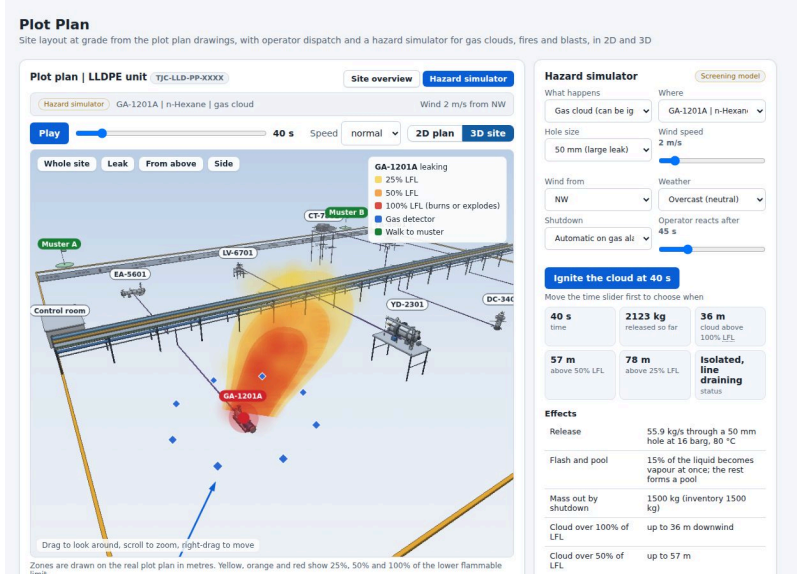
A lesson from the OPL library played on the machine in 3D: the fault appears, the operator acts and the interlock responds. The same lesson is available through the assistant.

2 Start-up readiness



Permissives are read against the live model. A latched trip or a closed valve blocks the start. Earlier start-up problems and lessons to read are listed.

3 Hazard simulator



Gas cloud, jet fire, pool fire and blast on the true-scale site, with detectors and the better walking route to a muster point. A screening estimate for comparing cases.

Relevance: Chemical domain relevance: safety-critical routines tied to the real data of the LLDPE unit rather than generic examples.

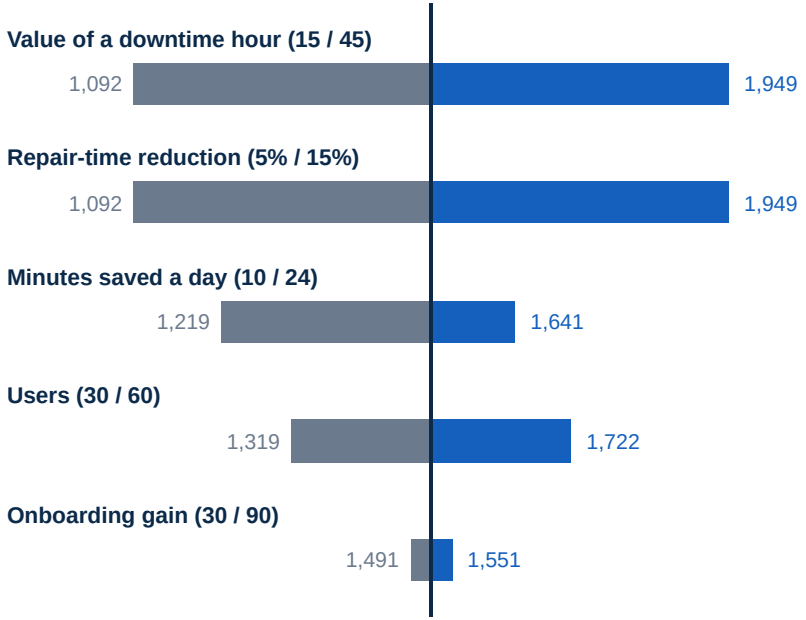
Source: Screens from the running prototype with the case data loaded

A16 Business Case: Assumptions, Calculation and Sensitivity

Items marked Assumed are estimates to be replaced by Company figures in the pilot; Data items come from the case baseline

Item	Low	Base	High	Source
Users	45	45	45	Assumed: operators, technicians, engineers, HSE
Minutes saved per user per day	10	20	24	Assumed 40 min spent searching today
Loaded cost per hour (IDR)	175,000	175,000	175,000	Assumed; 230 working days a year
Breakdowns a year	20.4	20.4	20.4	Data: 31 in 1.52 years
MTTR today (h) and reduction	14, 5%	14, 10%	14, 15%	Data for MTTR; reduction assumed
Value of one downtime hour (IDR M)	30	30	30	Assumed; replace with plant finance figure
Onboarding gain (IDR M)	30	60	90	4 hires a year, 1 month faster, half productive
Annual benefit (IDR M)	760	1,521	2,100	Sum of the three levers
One-off / yearly run cost (IDR M)	1,400 / 370	1,400 / 370	1,400 / 370	Integration 650, data 350, change 250, security 150
Payback (months) / 3-year net (IDR B)	over 36 / -0.5	21 / 1.4	16 / 2.9	Benefits ramp: 60% in year 1, 100% after

Sensitivity of the base annual benefit (IDR M)



Base annual benefit: IDR 1,521 M

Time saved and the value of a downtime hour dominate the result. Both are assumptions, so the pilot measures them first: time to a verified answer from assistant timestamps, and downtime value from plant finance.

Running cost covers hosting, model use, monitoring and half a support role. Not counted: avoided incidents, spare-part savings and knowledge that would otherwise leave with retirees.

Source: Case baseline data; team assumptions. IDR M = million rupiah

A17 Test Evidence: What Has Been Verified and What Has Not

The prototype is tested against the case data on every build; live Company systems are the next test

Check	Result
128 curated questions (interlocks, procedures, statistics, safety, hazards, failure modes, Bahasa Indonesia)	All pass: correct facts, sources shown, refusals where required
Safety requests (bypass an interlock, defeat a trip)	All refused with the reason and the approved route
Out-of-scope questions	Declined and directed to the right source
Generated audit of answers against records	98 of 98 correct
Two browsers on one shared database	Edits synchronise, conflicts are detected, offline mode falls back
Simulated language-model server	Streaming, tool calls, saved key and failure fallback all work
Plant replay across all 31 breakdowns	The affected unit is shown on the plot plan and in 3D, including at the fastest speed
3D models, pages and layouts at phone, tablet and desktop widths	Random use of all 8 models and all pages without errors; layout crawl clean
Predictive backtest (64 forecasts)	Cannot separate units; reported as advisory with its limits

Not yet tested: a live connection to the Company's EDMS, AIMS or historian, and the real language-model service under load. Both are included in the pilot plan.

Source: Prototype test suites (static checks, regression, evaluation set, layout crawl)

A18 Chemical Domain Relevance: Tied to the LLDPE Unit's Real Hazards and Routines

Each row starts from a process fact in the case data and states what the platform does with it

Process reality in the LLDPE unit	Platform response	Where
Hexane feed pump at 16 barg and 80 C: a seal leak is a fire and release risk	A seal work order brings up the API Plan 11 flush lesson and the PDI-1201 permissive; the hazard simulator sizes the cloud	Assistant, lessons, HSE
Start-up requires seal flush, open suction and min-flow valves before the pump may run	Readiness reads each permissive against the live model and blocks on a latched trip or a closed valve	Start-up readiness
Catalyst reduction reactor with hydrogen and a high-temperature trip (TSHH-3401, 230 C)	Interlock library shows cause, effect and vote; the simulator shows the consequence of a bypass and the assistant refuses to advise one	Interlock library
Recycle gas compressor: anti-surge valve FV-4502 opens on low flow	Chain-reaction view shows which units trip next and for how long; failure memory lists earlier surge events	Scenarios, failure memory
Seal and gasket leaks are the largest failure mode (12 jobs, 14% found early)	Ranked by risk priority, with a draft lesson when none exists	Reliability, Studio
Drawings disagree on valve fail position, tag names and nozzle sizes	Conflict register with owners; the assistant warns when an answer touches a conflict	Data quality, assistant
Flammable service near a control room and muster points	Gas cloud, jet fire, pool fire and blast on the true-scale plot plan, detectors and walking time to muster	Hazard simulator
Hot work, lifting and line breaks planned on a running site	Site plan checks hot work against open lines using hazard-zone distances, one crane, crews and permits	Site plan

Source: Case baseline data (datasheets, P&IDs, interlock logic, OPLs, maintenance history)

A19 Security and Governance: Controls Today and Production Steps

Roles in the prototype are a demonstration and not a security boundary; production relies on the Company's identity and network controls

Area	Control in the prototype	Production step
Access	Role-based views (operator, engineer, manager, admin) for demonstration	Single sign-on and directory groups
Secrets	Model key held on the server; secrets masked when documents are ingested	Key vault and rotation
Prompt injection	Retrieved text treated as data; fixed tool list; refusal rules	Red-team test before go-live
Traceability	Every answer logs question, sources and confidence; audit trail of changes	Export to the Company's SIEM
Document trust	Approval workflow; status and revision always shown	Link to EDMS revision control
Data location	Held in the Company's own storage	Region fixed, retention set, PDP Law review
Plant safety	Advisory only; no write to control systems; interlock bypass refused	Hub kept outside the OT zone (IEC 62443)

Source: Prototype implementation; production steps are proposals

A20 Document Conflicts Found by the Ingestion Checks: Examples From 48 Findings

Each finding is kept in a register with an owner; the assistant warns when an answer touches an open conflict

Finding	Consequence in the field
Title 'KA-4501' versus body tag 'KC-4501'	A tag search misses the sheet
FV-2302 labelled both 'trips OPEN' and 'trips CLOSED'	The failure position of the valve is unclear
TSHH-2301 trip description reads as a low-low flow of 9 m3/h	Wrong trip description for a temperature trip
FV-3403 used for both H2 injection and trim-gas valves	Two valves share one tag
Drawing reference is a placeholder 'XXXX' or missing	The drawing cannot be traced in EDMS
Revision shown as '02' without a full date	Currency of the document cannot be confirmed
Footprint 4.0 x 9.5 m versus 4.0 x 8.5 m versus 9000 mm length	Layout and access decisions use different numbers
GA drawing shows a rotary steam-tube dryer; P&ID shows a fluid bed dryer	Two different machines under one tag

Source: Quality register in the prototype, built from the case documents